Example 5.18

The following table shows age distribution of persons in a particular region:

Age (years)	No. of persons (in thousands)
Below 10	2
Below20	5
Below30	9
Below 40	12
Below 50	14
Below 60	15
Below 70	15.5
Below 80	15.6

Find the median age.

Solution:

We are given upper limit and less than cumulative frequencies. First find the class-intervals and the frequencies. Since the values are increasing by 10, hence the width of the class interval is equal to 10.

Age groups	No. of persons (in thousands) f	cf
0 – 10	2	2
10 – 20	3	5
20 – 30	4	9
30 – 40	3	12
40 – 50	2	14
50 – 60	1	15
60 – 70	0.5	15.5
70 – 80	0.1	15.6
Total	**N = 15.6**	

$$\left(\frac{N}{2}\right) = \frac{15.6}{2} = 7.8$$

Median lies in the 20 – 30 age group

Ch5_11th Stats.indd 130 2/3/2019 11:00:59 AM